

LOB Event-Stream Models — Evaluation Report

Prediction & closed-loop simulation of 7 marked temporal point processes on Gemini-ETH

1 Setup

Seven models trained and evaluated under **identical** settings, so prediction and simulation are directly comparable. Baselines: **Hawkes**, **LSTM**, **SAHP**, **CT-LSTM**, **PCT-LSTM**, **S2P2**; proposed: **SS2P2** (S2P2 backbone + gated-bounded G1 rate $\times$ rate-neutral marks).

Data	<code>gmni_eth_v2_marks</code> — Gemini ETH, 7 days, 100% single-item ; real rate ≈ 2.32 ev/s
Train	seq 64 / stride 32, 40 epochs, batch 256, lr 2×10^{-3} , categorical mark head, hidden 64
Prediction	per genuine event; <code>genuine_eval</code> (head-agnostic)
Simulation	closed-loop rollout (inversion sampler, same for all), Cont (2001) stylized facts
Hardware	1 $\times$ NVIDIA A40 per model; pipeline <code>scripts/run_eval_all.sh</code>

2 Prediction (per genuine event)

overall NLL (time + mark) is the headline fit gate; *mean u* is the time-rescaling calibration ($\rightarrow 1$ if the intensity is correct); KS is the Kolmogorov–Smirnov distance of the compensator masses from $\text{Exp}(1)$; tMAE is next-time error; ACC/PPL are next-type accuracy/perplexity.

model	overall $\downarrow$	timeNLL	markNLL	KS $\downarrow$	mean u ($\rightarrow 1$)	tMAE(s)	ACC $\uparrow$	PPL $\downarrow$
SS2P2	1.078	−1.219	2.297	0.253	2.09	0.344	0.343	9.95
Hawkes	1.192	−1.444	2.636	0.241	1.98	0.315	0.262	13.96
CT-LSTM	1.192	−1.444	2.636	0.241	1.98	0.315	0.262	13.96
PCT-LSTM	1.327	−1.214	2.541	0.189	1.87	0.397	0.276	12.70
LSTM	2.573	−0.127	2.700	0.494	1.43	0.516	0.240	14.87
SAHP	2.811	0.083	2.728	0.496	1.60	0.532	0.236	15.31
S2P2	2.959	0.613	2.346	0.437	4.29	1.430	0.326	10.45

SS2P2 wins all three headline metrics (overall-NLL, ACC, PPL). S2P2 has the 2nd-best ACC/PPL but the *worst* overall-NLL — its timing is badly miscalibrated (mean $u = 4.29$). Hawkes $\equiv$ CT-LSTM (same neural-Hawkes decoder).

3 Simulation — stylized-facts fit set

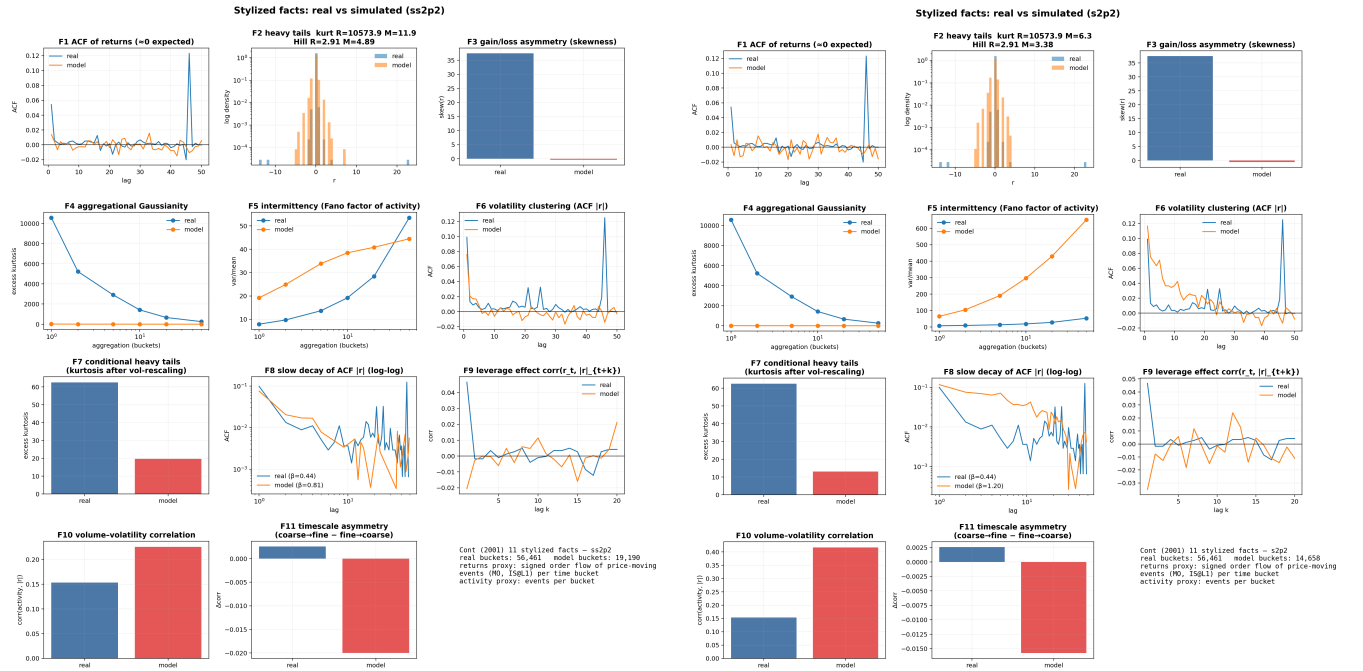
Relative error vs. real (lower better). Mean event rate is the precondition: if the simulated rate is wrong, the other facts are computed on a mis-scaled stream.

model	sim rate	rate_re↓	Fano_re↓	clus_re↓	retACF_re↓	kurt_re
SAHP	2.23	0.04	0.345	0.237	0.353	≈ 1.0
LSTM	1.92	0.17	0.276	0.104	0.508	≈ 1.0
SS2P2	52.7	21.7	1.015	0.215	0.401	≈ 1.0
PCT-LSTM	23.6	9.2	0.335	0.319	1.114	≈ 1.0
Hawkes/CT-LSTM	54.5	22.5	4.90	0.044	0.293	≈ 1.0
S2P2	87.3	36.6	11.63	2.596	0.002	≈ 1.0

SS2P2 is far more stable than its S2P2 parent: Fano 1.0 vs 11.6, clustering 0.22 vs 2.60, rate 53 vs 87 — the gated G1 bound contains the closed-loop rate that S2P2’s unbounded `softplus` rate lets run away. The simple LSTM/SAHP match the unconditional rate/Fano best, but they are the *weakest predictors* and carry no conditional dynamics. Excess kurtosis is unmatched by all (real 1-s kurtosis is extreme, $\sim 10^4$).

4 Reading

- **Best predictor:** SS2P2, simultaneously best on overall-NLL, accuracy and perplexity.
- **Best prediction/stability trade-off:** SS2P2 — it beats S2P2 on *both* prediction and simulation stability at once (the rate-neutral mark head keeps accuracy; the G1 bound keeps the rollout from exploding).
- **Cross-cutting limitation:** no intensity model is rate-calibrated free-running (mean $u > 1$, simulated rate $\gg$ real) — the windowed-training exposure-bias effect. Only the trivial LSTM/SAHP match rate, by virtue of weak dynamics. Fixing this (stateful / full-sequence training) is the clear next step.



Stylized facts (real blue vs. model red): **SS2P2 (left)** keeps Fano/clustering bounded and close; **S2P2 (right)** over-disperses and over-clusters as its rate runs away.

Pipeline scripts/run.eval.all.sh (job 7003907/7004334), experiments/eval.all/REPORT.txt, repo [honglinfu98/simulation](https://github.com/honglinfu98/simulation).